

Instructions:

You will work on these exercises during the practical session in the third week of the course. We will cover exercise 1. We encourage you to try it before the session.

Exercise 1: Naive Bayes

Naive Bayes is a classification model that makes strong independence assumptions about the features of the data, conditional on the class label. Specifically, it assumes that each feature is independent given the class label. In contrast, a full-covariance Gaussian class-conditional model can represent correlations between all input features.

If correlated features are treated independently, the evidence for a class may be overcounted. However, Naive Bayes is simple to construct because, by ignoring correlations, the class-conditional likelihood $p(\mathbf{x} | \mathcal{C}_k)$ factors into a product of D univariate distributions:

$$p(\mathbf{x} | \mathcal{C}_k) = \prod_{d=1}^D p(x_d | \mathcal{C}_k).$$

Consider a document classification task with K classes $\mathcal{C}_1, \dots, \mathcal{C}_K$. We represent each document using a bag-of-words vector $\mathbf{x}_n \in \{0, 1\}^D$, where $x_{nd} = 1$ if word d appears in document n , and $x_{nd} = 0$ otherwise. The training inputs form a matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$. The targets form a matrix $\mathbf{T} \in \mathbb{R}^{N \times K}$, whose n -th row is a one-hot vector $\mathbf{t}_n^\top = (t_{n1}, \dots, t_{nK})$ with $t_{nk} = 1$ if document n belongs to class $\mathcal{C}_k$.

Assume the class prior probabilities are known and given by

$$p(\mathcal{C}_k) = \pi_k, \quad \sum_{k=1}^K \pi_k = 1.$$

For this exercise, model word presence/absence using a Bernoulli distribution. Thus, conditioned on class $\mathcal{C}_k$, word d is Bernoulli distributed with parameter

$$\theta_{dk} = p(x_d = 1 | \mathcal{C}_k),$$

and

$$p(\mathbf{x} | \mathcal{C}_k, \theta_{1k}, \dots, \theta_{Dk}) = \prod_{d=1}^D \theta_{dk}^{x_d} (1 - \theta_{dk})^{1-x_d}.$$

- (a) Write down the data likelihood $p(\mathbf{T}, \mathbf{X} | \Theta)$ without first using the Naive Bayes independence assumption. Then derive the likelihood for the general K -class Naive Bayes classifier, stating where you use the product rule and the Naive Bayes assumption. Write the likelihood in terms of $p(x_d | \mathcal{C}_k)$, without assuming the explicit Bernoulli form.

Solution:

Using the IID assumption over documents,

$$p(\mathbf{T}, \mathbf{X} | \Theta) = \prod_{n=1}^N p(\mathbf{t}_n, \mathbf{x}_n | \Theta).$$

By the product rule,

$$p(\mathbf{t}_n, \mathbf{x}_n | \Theta) = p(\mathbf{t}_n | \Theta) p(\mathbf{x}_n | \mathbf{t}_n, \Theta).$$

Since $\mathbf{t}_n$ is one-hot, the class label can be selected by the entries t_{nk} :

$$p(\mathbf{T}, \mathbf{X} \mid \Theta) = \prod_{n=1}^N \prod_{k=1}^K [p(\mathcal{C}_k) p(\mathbf{x}_n \mid \mathcal{C}_k, \Theta)]^{t_{nk}}.$$

Using $p(\mathcal{C}_k) = \pi_k$, this becomes

$$p(\mathbf{T}, \mathbf{X} \mid \Theta) = \prod_{n=1}^N \prod_{k=1}^K [\pi_k p(\mathbf{x}_n \mid \mathcal{C}_k, \Theta)]^{t_{nk}}.$$

Finally, the Naive Bayes assumption gives

$$p(\mathbf{x}_n \mid \mathcal{C}_k, \Theta) = \prod_{d=1}^D p(x_{nd} \mid \mathcal{C}_k, \theta_{dk}).$$

Therefore,

$$p(\mathbf{T}, \mathbf{X} \mid \Theta) = \prod_{n=1}^N \prod_{k=1}^K \left[\pi_k \prod_{d=1}^D p(x_{nd} \mid \mathcal{C}_k, \theta_{dk}) \right]^{t_{nk}}.$$

(b) Write down the data log-likelihood $\ln p(\mathbf{T}, \mathbf{X} \mid \Theta)$ for the Bernoulli model.

Solution:

For the Bernoulli model,

$$p(x_{nd} \mid \mathcal{C}_k, \theta_{dk}) = \theta_{dk}^{x_{nd}} (1 - \theta_{dk})^{1-x_{nd}}.$$

Substituting this into the Naive Bayes likelihood gives

$$p(\mathbf{T}, \mathbf{X} \mid \Theta) = \prod_{n=1}^N \prod_{k=1}^K \left[\pi_k \prod_{d=1}^D \theta_{dk}^{x_{nd}} (1 - \theta_{dk})^{1-x_{nd}} \right]^{t_{nk}}.$$

Taking logarithms,

$$\ln p(\mathbf{T}, \mathbf{X} \mid \Theta) = \sum_{n=1}^N \sum_{k=1}^K t_{nk} \left[\ln \pi_k + \sum_{d=1}^D (x_{nd} \ln \theta_{dk} + (1 - x_{nd}) \ln(1 - \theta_{dk})) \right].$$

(c) Solve for the maximum likelihood estimator of θ_{dk} . Explain how the result can be interpreted.

Solution:

Keep only the terms in the log-likelihood that depend on a fixed θ_{dk} :

$$\ell(\theta_{dk}) = \sum_{n=1}^N t_{nk} [x_{nd} \ln \theta_{dk} + (1 - x_{nd}) \ln(1 - \theta_{dk})] + C.$$

Differentiating and setting the derivative equal to zero gives

$$\frac{\partial \ell}{\partial \theta_{dk}} = \sum_{n=1}^N t_{nk} \left(\frac{x_{nd}}{\theta_{dk}} - \frac{1 - x_{nd}}{1 - \theta_{dk}} \right) = 0.$$

Combining the terms inside the sum,

$$\sum_{n=1}^N t_{nk} \frac{x_{nd} - \theta_{dk}}{\theta_{dk}(1 - \theta_{dk})} = 0.$$

Assuming $0 < \theta_{dk} < 1$, multiply by $\theta_{dk}(1 - \theta_{dk})$:

$$\sum_{n=1}^N t_{nk}(x_{nd} - \theta_{dk}) = 0.$$

Hence

$$\sum_{n=1}^N t_{nk}x_{nd} = \theta_{dk} \sum_{n=1}^N t_{nk}.$$

Define

$$N_k = \sum_{n=1}^N t_{nk}, \quad N_{dk} = \sum_{n=1}^N t_{nk}x_{nd}.$$

Then

$$\hat{\theta}_{dk} = \frac{N_{dk}}{N_k} = \frac{\sum_{n=1}^N t_{nk}x_{nd}}{\sum_{n=1}^N t_{nk}}.$$

Thus, $\hat{\theta}_{dk}$ is the fraction of documents in class $\mathcal{C}_k$ that contain word d .

(d) Write $p(\mathcal{C}_1 | \mathbf{x})$ for the general K -class Naive Bayes classifier.

Solution:

By Bayes' rule,

$$p(\mathcal{C}_1 | \mathbf{x}) = \frac{p(\mathbf{x} | \mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x})}.$$

Marginalizing over all classes gives

$$p(\mathbf{x}) = \sum_{k=1}^K p(\mathbf{x} | \mathcal{C}_k)p(\mathcal{C}_k).$$

Therefore,

$$p(\mathcal{C}_1 | \mathbf{x}) = \frac{\pi_1 \prod_{d=1}^D p(x_d | \mathcal{C}_1, \theta_{d1})}{\sum_{k=1}^K \pi_k \prod_{d=1}^D p(x_d | \mathcal{C}_k, \theta_{dk})}.$$

(e) Write $p(\mathcal{C}_1 | \mathbf{x})$ for the Bernoulli model.

Solution:

Substitute the Bernoulli class-conditional likelihood

$$p(x_d | \mathcal{C}_k, \theta_{dk}) = \theta_{dk}^{x_d} (1 - \theta_{dk})^{1-x_d}$$

into the previous result. This gives

$$p(\mathcal{C}_1 | \mathbf{x}) = \frac{\pi_1 \prod_{d=1}^D \theta_{d1}^{x_d} (1 - \theta_{d1})^{1-x_d}}{\sum_{k=1}^K \pi_k \prod_{d=1}^D \theta_{dk}^{x_d} (1 - \theta_{dk})^{1-x_d}}.$$

Exercise 2: LDA and QDA

Exam question

Suppose we have a dataset consisting of data points with two features, with each data point assigned to one of two classes, $\mathcal{C}_1$ or $\mathcal{C}_2$. Suppose that the class-conditional likelihoods are known:

$$p(\mathbf{x} | \mathcal{C}_1) = \mathcal{N}\left(\mathbf{x} \mid \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}\right), \quad p(\mathbf{x} | \mathcal{C}_2) = \mathcal{N}\left(\mathbf{x} \mid \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right).$$

- (a) Were we to perform posterior classification using this model, would it fall in the category of Linear Discriminant Analysis (LDA) or Quadratic Discriminant Analysis (QDA)? Why?

Solution:

QDA is more appropriate. LDA assumes that all classes share the same covariance matrix, which leads to linear decision boundaries. Here the two classes have different covariance matrices, so the decision boundary is quadratic. QDA allows class-specific covariance matrices and can represent this boundary.

- (b) Assume we are told, before seeing any data, that we should expect twice as many data points in $\mathcal{C}_2$ as in $\mathcal{C}_1$. Write out the posteriors $p(\mathcal{C}_1 | \mathbf{x})$ and $p(\mathcal{C}_2 | \mathbf{x})$ given this information. You can leave the evidence term $p(\mathbf{x})$ in your expressions.

Recall that the multivariate Gaussian density is

$$\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right].$$

Solution:

The statement about class frequencies gives

$$p(\mathcal{C}_2) = 2p(\mathcal{C}_1), \quad p(\mathcal{C}_1) + p(\mathcal{C}_2) = 1.$$

Therefore

$$p(\mathcal{C}_1) = \frac{1}{3}, \quad p(\mathcal{C}_2) = \frac{2}{3}.$$

For class $\mathcal{C}_1$, we have $|4\mathbf{I}| = 16$ and $(4\mathbf{I})^{-1} = \frac{1}{4}\mathbf{I}$, so

$$p(\mathbf{x} | \mathcal{C}_1) = \frac{1}{2\pi} \frac{1}{4} \exp \left(-\frac{1}{8} \mathbf{x}^\top \mathbf{x} \right) = \frac{1}{8\pi} \exp \left(-\frac{1}{8} \mathbf{x}^\top \mathbf{x} \right).$$

For class $\mathcal{C}_2$, we have $|\mathbf{I}| = 1$ and $\mathbf{I}^{-1} = \mathbf{I}$, so

$$p(\mathbf{x} | \mathcal{C}_2) = \frac{1}{2\pi} \exp \left(-\frac{1}{2} \mathbf{x}^\top \mathbf{x} \right).$$

Applying Bayes' rule gives

$$p(\mathcal{C}_1 | \mathbf{x}) = \frac{p(\mathcal{C}_1)p(\mathbf{x} | \mathcal{C}_1)}{p(\mathbf{x})} = \frac{\frac{1}{3} \frac{1}{8\pi} \exp \left(-\frac{1}{8} \mathbf{x}^\top \mathbf{x} \right)}{p(\mathbf{x})},$$

and

$$p(\mathcal{C}_2 | \mathbf{x}) = \frac{p(\mathcal{C}_2)p(\mathbf{x} | \mathcal{C}_2)}{p(\mathbf{x})} = \frac{\frac{2}{3} \frac{1}{2\pi} \exp \left(-\frac{1}{2} \mathbf{x}^\top \mathbf{x} \right)}{p(\mathbf{x})}.$$

- (c) Come up with a condition of the form $\mathbf{x}^\top \mathbf{x} > a$ defining when we classify $\mathbf{x}$ as $\mathcal{C}_1$ rather than $\mathcal{C}_2$.

Solution:

We classify $\mathbf{x}$ as $\mathcal{C}_1$ when

$$p(\mathcal{C}_1 | \mathbf{x}) > p(\mathcal{C}_2 | \mathbf{x}).$$

Using the posteriors from the previous part, this is equivalent to

$$\frac{1}{24\pi} \exp\left(-\frac{1}{8}\mathbf{x}^\top \mathbf{x}\right) > \frac{1}{3\pi} \exp\left(-\frac{1}{2}\mathbf{x}^\top \mathbf{x}\right),$$

since the evidence $p(\mathbf{x})$ is positive and appears on both sides. Rearranging,

$$\exp\left(-\frac{1}{8}\mathbf{x}^\top \mathbf{x}\right) > 8 \exp\left(-\frac{1}{2}\mathbf{x}^\top \mathbf{x}\right).$$

Taking logarithms gives

$$-\frac{1}{8}\mathbf{x}^\top \mathbf{x} > \ln 8 - \frac{1}{2}\mathbf{x}^\top \mathbf{x}.$$

Therefore,

$$\frac{3}{8}\mathbf{x}^\top \mathbf{x} > \ln 8,$$

and the classification rule is

$$\mathbf{x}^\top \mathbf{x} > \frac{8 \ln 8}{3}.$$

Points satisfying this condition are classified as $\mathcal{C}_1$; otherwise, they are classified as $\mathcal{C}_2$.

Exercise 3: Classification regions

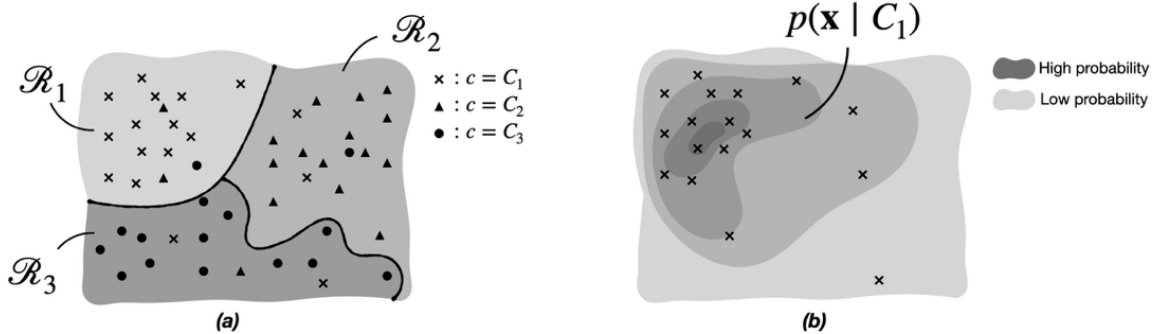
Exam question

Consider a three-class classification problem with points $\mathbf{x} \in \mathbb{R}^2$ that belong to one of three classes $\mathcal{C}_1$, $\mathcal{C}_2$, or $\mathcal{C}_3$. We build a classifier by partitioning the 2D plane into three decision regions $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$, corresponding to the three classes. That is, we classify a point $\mathbf{x}$ as class $\mathcal{C}_k$ when $\mathbf{x} \in \mathcal{R}_k$.

We assume that the observations follow class-conditional distributions

$$\mathbf{x} \sim p(\mathbf{x} | c = \mathcal{C}_k),$$

which we write more compactly as $p(\mathbf{x} | \mathcal{C}_k)$.



- (a) Suppose we know the class-conditional distribution $p(\mathbf{x} | \mathcal{C}_1)$. Give the probability that a point sampled from this distribution is correctly classified as class $\mathcal{C}_1$.

Solution:

A point from class $\mathcal{C}_1$ is classified correctly exactly when it falls in region $\mathcal{R}_1$. Therefore,

$$p(\mathbf{x} \in \mathcal{R}_1 \mid \mathcal{C}_1) = \int_{\mathcal{R}_1} p(\mathbf{x} \mid \mathcal{C}_1) d\mathbf{x}.$$

- (b) Without knowing anything about the class-conditional distributions of the other classes, define the total error rate that the classifier makes on class $\mathcal{C}_1$. Let the total error rate be the fraction of times a point $\mathbf{x}$ is assigned any class other than its true class. Write down the expected error rate for points in class $\mathcal{C}_1$.

Solution:

The probability of a correct prediction for class $\mathcal{C}_1$ is

$$p(\mathbf{x} \in \mathcal{R}_1 \mid \mathcal{C}_1).$$

Hence the expected error rate on class $\mathcal{C}_1$ is

$$1 - p(\mathbf{x} \in \mathcal{R}_1 \mid \mathcal{C}_1) = 1 - \int_{\mathcal{R}_1} p(\mathbf{x} \mid \mathcal{C}_1) d\mathbf{x}.$$

Equivalently,

$$p(\mathbf{x} \in \mathcal{R}_2 \cup \mathcal{R}_3 \mid \mathcal{C}_1).$$

- (c) All forms of errors, and the frequencies with which they are expected to occur, can be characterized by an expected confusion matrix. Define the expected confusion matrix in terms of the three class-conditional distributions and the decision regions $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$.

Solution:

Let rows correspond to predicted classes and columns correspond to true classes. The expected confusion matrix is

	$\mathcal{C}_1$	$\mathcal{C}_2$	$\mathcal{C}_3$
$\mathcal{R}_1$	$p(\mathbf{x} \in \mathcal{R}_1 \mid \mathcal{C}_1)$	$p(\mathbf{x} \in \mathcal{R}_1 \mid \mathcal{C}_2)$	$p(\mathbf{x} \in \mathcal{R}_1 \mid \mathcal{C}_3)$
$\mathcal{R}_2$	$p(\mathbf{x} \in \mathcal{R}_2 \mid \mathcal{C}_1)$	$p(\mathbf{x} \in \mathcal{R}_2 \mid \mathcal{C}_2)$	$p(\mathbf{x} \in \mathcal{R}_2 \mid \mathcal{C}_3)$
$\mathcal{R}_3$	$p(\mathbf{x} \in \mathcal{R}_3 \mid \mathcal{C}_1)$	$p(\mathbf{x} \in \mathcal{R}_3 \mid \mathcal{C}_2)$	$p(\mathbf{x} \in \mathcal{R}_3 \mid \mathcal{C}_3)$

More compactly, entry (k, l) is

$$p(\mathbf{x} \in \mathcal{R}_k \mid \mathcal{C}_l) = \int_{\mathcal{R}_k} p(\mathbf{x} \mid \mathcal{C}_l) d\mathbf{x}.$$

- (d) For the purpose of classifying points into the three classes, instead of checking in which region $\mathcal{R}_k$ a point falls, one could evaluate the posterior probability that the point belongs to class $\mathcal{C}_k$, assuming that the class priors $p(\mathcal{C}_k)$ are known. Give the expression for $p(\mathcal{C}_k \mid \mathbf{x})$.

Solution:

By Bayes' rule,

$$p(\mathcal{C}_k \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \mathcal{C}_k)p(\mathcal{C}_k)}{p(\mathbf{x})} = \frac{p(\mathbf{x} \mid \mathcal{C}_k)p(\mathcal{C}_k)}{\sum_{l=1}^3 p(\mathbf{x} \mid \mathcal{C}_l)p(\mathcal{C}_l)}.$$

- (e) Suppose we know all true class-conditional distributions. How would you define optimal decision regions? The best possible decision region $\mathcal{R}_i$ for class $\mathcal{C}_i$ is one such that what condition holds?

Solution:

The optimal decision rule assigns each point to the class with largest posterior probability. Thus,

$$\mathcal{R}_i = \{\mathbf{x} : p(\mathcal{C}_i | \mathbf{x}) \geq p(\mathcal{C}_k | \mathbf{x}) \text{ for all } k \neq i\}.$$

Equivalently, using Bayes' rule, this can be written as

$$\mathcal{R}_i = \{\mathbf{x} : p(\mathbf{x} | \mathcal{C}_i)p(\mathcal{C}_i) \geq p(\mathbf{x} | \mathcal{C}_k)p(\mathcal{C}_k) \text{ for all } k \neq i\}.$$

- (f) Now assume that the data are sampled from the following probabilistic model. Each class-conditional distribution is Gaussian with mean $\boldsymbol{\mu}_k$ and covariance matrix $\boldsymbol{\Sigma}_k$,

$$p(\mathbf{x} | \mathcal{C}_k) = \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),$$

and the prior probability of class $\mathcal{C}_k$ is $p(\mathcal{C}_k) = \pi_k$, where $\sum_{k=1}^3 \pi_k = 1$. Consider a dataset

$$\mathcal{D} = \{(\mathbf{x}_n, \mathcal{C}_n)\}_{n=1}^N,$$

and define the one-hot variable $z_{nk} = 1$ if $\mathcal{C}_n = \mathcal{C}_k$ and $z_{nk} = 0$ otherwise. Write down the likelihood of observing $\mathcal{D}$ under the parameters $\{\boldsymbol{\mu}_k\}$, $\{\boldsymbol{\Sigma}_k\}$, and $\{\pi_k\}$.

Solution:

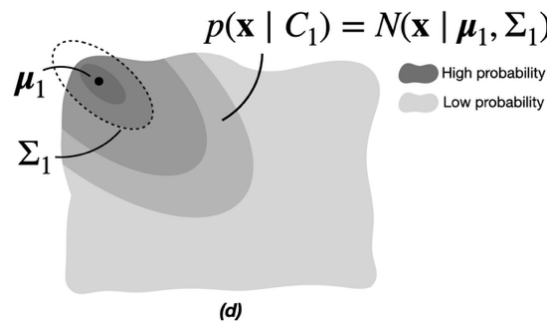
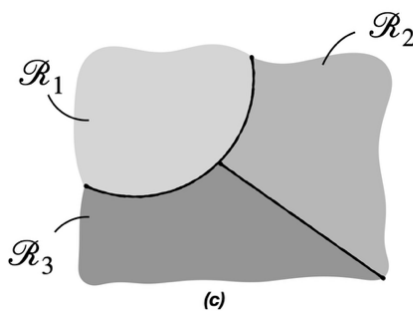
For one observation,

$$p(\mathbf{x}_n, \mathcal{C}_n | \{\boldsymbol{\mu}_k\}, \{\boldsymbol{\Sigma}_k\}, \{\pi_k\}) = \prod_{k=1}^3 [\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)]^{z_{nk}}.$$

Assuming the observations are independent, the likelihood is

$$p(\mathcal{D} | \{\boldsymbol{\mu}_k\}, \{\boldsymbol{\Sigma}_k\}, \{\pi_k\}) = \prod_{n=1}^N \prod_{k=1}^3 [\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)]^{z_{nk}}.$$

- (g) Suppose the optimal decision boundaries are known, as illustrated below. Looking at the decision boundaries, what can we say about the covariance matrices?



Solution:

The boundary between $\mathcal{C}_2$ and $\mathcal{C}_3$ is linear, which suggests that these two classes have the

same covariance matrix:

$$\Sigma_2 = \Sigma_3.$$

The boundaries involving $\mathcal{C}_1$ are curved, so the covariance matrix for class $\mathcal{C}_1$ differs from the others:

$$\Sigma_1 \neq \Sigma_2, \quad \Sigma_1 \neq \Sigma_3.$$